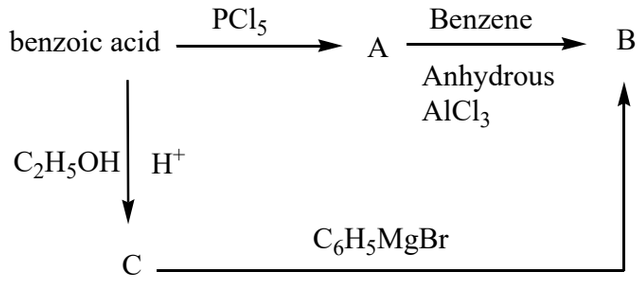
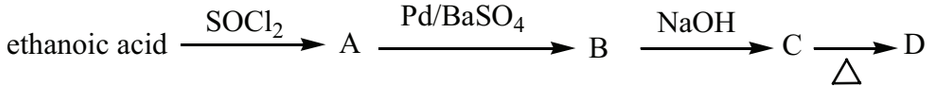
4. Identify A, B and C



5. Identify A, B, C and D



6. An alkene (A) on ozonolysis gives propanone and aldehyde (B). When (B) is oxidised (C) is obtained. (C) is treated with Br_2/P gives (D) which on hydrolysis gives (E). When propanone is treated with HCN followed by hydrolysis gives (E). Identify A, B, C, D and E.
7. How will you convert benzaldehyde into the following compounds?

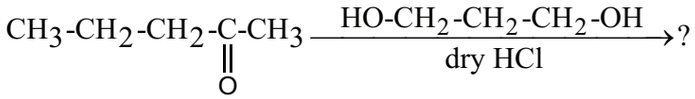
   (i) benzophenone (ii) benzoic acid

   (iii) α-hydroxyphenylaceticacid.
8. What is the action of HCN on

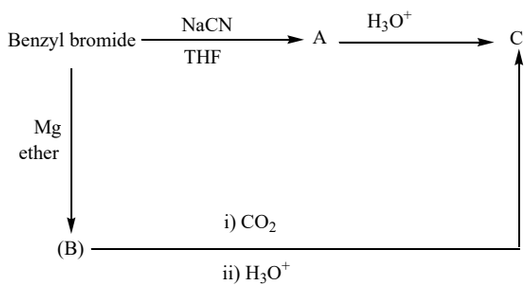
   (i) propanone (ii) 2,4-dichlorobenzaldehyde. iii) ethanal
9. A carbonyl compound A having molecular formula $C_5H_{10}O$ forms crystalline precipitate with sodium bisulphite and gives positive iodoform test. A does not reduce Fehling solution. Identify A.
10. Write the structure of the major product of the aldol condensation of benzaldehyde with acetone.
11. How are the following conversions effected

    (a) propanal into butanone (b) Hex-3-yne into hexan-3-one.

    (c) phenylmethanal into benzoic acid (d) phenylmethanal into benzoin
12. Complete the following reaction.



13. Identify A, B and C



XII U12-Carbonyl Compounds.indd 193 08-12-2021 22:50:55